Estimating in Building Construction

Chapter 9 Excavation

Excavation

- Type of soil
- Required slope
- Bracing or Sheet piling
- Groundwater / pumping required
- Excavation
  - be left on site
  - or must be removed
  - how far must be hauled
- Top soil be stock piled for future use

Calculating Excavation

- Determined from the site plan
  - Scaled in feet (no inches)
- Measured in cubic yards

Example 9-1 Required accuracy

Length = 52.83 ft

Width = 75.75 ft

Depth = 6.33 ft

Volume = 52.83 x 75.75 x 6.33

=25,331.85 cf

= 25,332/27

= 938 cy

Swell and Shrinkage

- Excavation measured in cubic yards
  - Must separate bank, loose, or compacted
- Bank: soils in original state (bcy)
- Loose: soils after excavation and before compaction (lcy)
- Compacted: Soils after placement and compaction (ccy)

Swell and Shrinkage

- Swell: increase in volume between bank and loose condition.
  - Expressed as percentage gained above the original volume.
- Shrinkage: decrease in volume between bank and compacted condition
  - Expressed as a percentage of the bank volume

Swell and Shrinkage

Percentage of Swell & Shrinkage		
Material	**Swell**	**Shrinkage**
Sand and Gravel	10 to 18%	85 to 100%
Loam (sand, silt, clay)	15 to 25%	90 to 100%
Dense Clay	20 to 35%	90 to 100%
Solid Rock	40 to 70%	130%
Solid rock when compacted is less dense than its bank condition.		

Example 9.2 Determining Swell and Haul

Dense clay = 1,000 bcy

Swell = 30%

Loads to be hauled = ?

Lcy = bcy x(1+swell)

Cubic yard of haul = 1,000 (1+30%)

= 1,300 lcy

Dump truck = 7cy

Load = 1,300 lcy / 7 lcy per load

= 185.71 = 186 loads

Example 9.3 Determining Shrinkage and Haul

Required compacted sand/gravel = 500 ccy

Swell of the material= 15%

Shrinkage of the material = 95%

Loads to be hauled = ?

Required bcy = required ccy / Shrinkage

= 500/0.95 = 526 bcy

Loose cubic yard of haul = 526 bcy (1+15%)

= 605 lcy

Dump truck used = 7cy

Load = 605 lcy / 7 lcy per load

= 86 loads

Equipment

- Trenching machine
- Bulldozer
- Power shovels
- Scrapers
- Loader
- Backhoe
- Clamshell

Earthwork- New grades

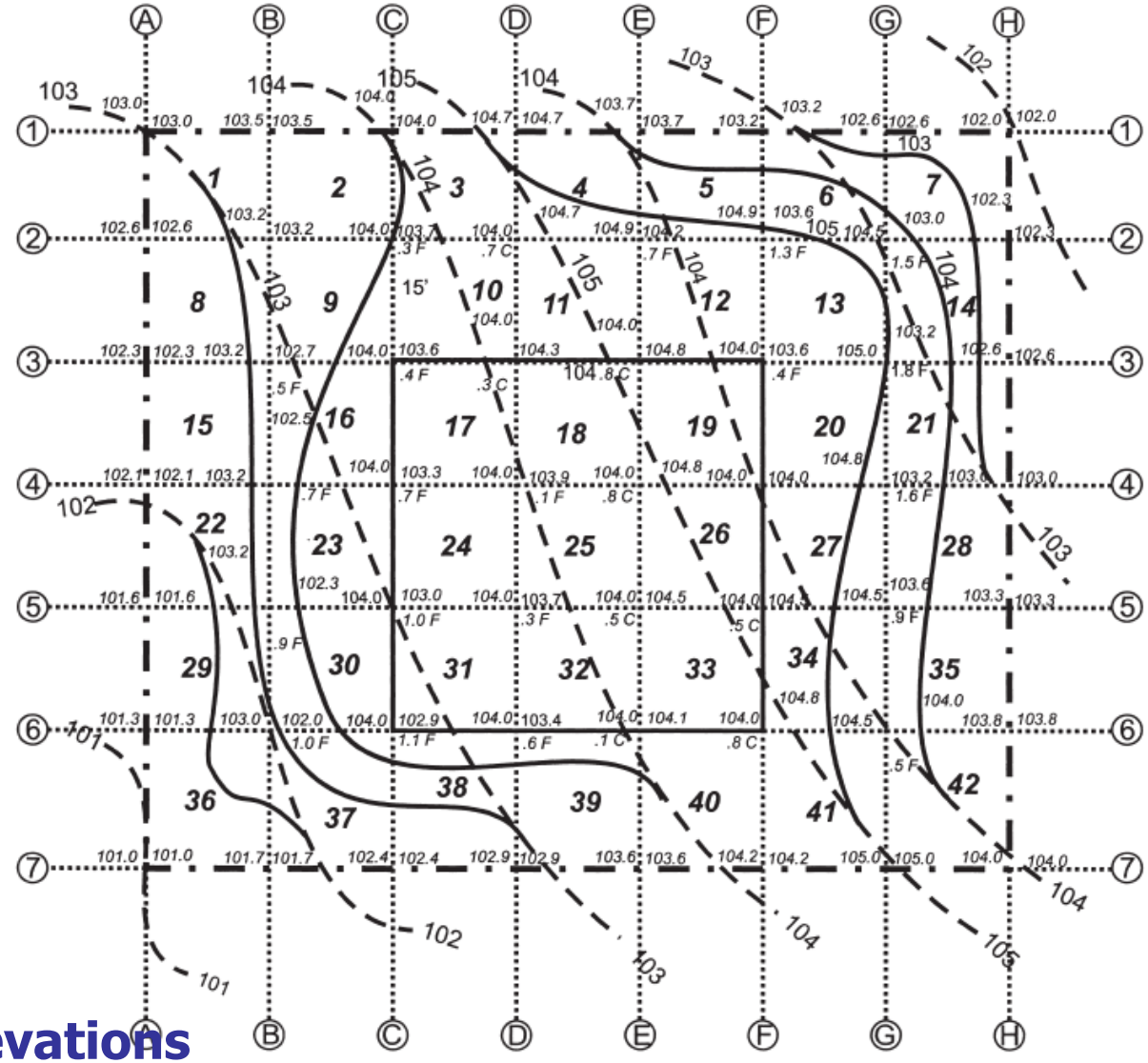
- Contour lines connect points of equal elevation
- Existing elevation are shown with dashed contour lines while proposed new elevations are denoted with solid lines.

Cut and Fill

- Cut: removal of soil to meet required grade
- Fill: addition of soil to meet required grade

Cross-Section Method

- Divide the site into grids (See Fig 9.3 and Fig. 9.4). Assign a no. to each grid.
- If the horizontal lines are numeric, the vertical lines would be alphabetic. The opposite is also true.
- At intersection of grid lines, determine:
  - Existing grade
  - Desired grade
  - Cut or fill



Grid with Elevations

Figure 9.6

Cross-Section Method

- Calculate cut or fill for each area

- $\text{Cut} = \dfrac{\text{Sum of cuts} \times \text{Area}}{\text{Number of cuts}}$

- $\text{Fills} = \dfrac{\text{Sum of fills} \times \text{Area}}{\text{Number of fills}}$

Example 9.4 Fill Volume

Using grid 13 from Fig 9.7, Determine the fill quantity

Point	Planned Elevation	Existing Elevation	Fill (ft)
F2	104.9′	103.6′	1.3
G2	104.5′	103.0′	1.5
F3	104.0′	103.6′	0.4
G3	105.0′	103.2′	1.8

ccf of fill = Sum of fill at intersections/Number of intersections x Area
= (1.3 + 1.5 + 0.4 +1.8) / 4 x 2500
= 3125 ccf

This amount is then entered in the fill column of the cut and fill worksheet (Figure 9.16)

Example 9.5 Cut Volume

Using Grid 40 from Fig 9.10, Determine the cut quantity

Point	Planned Elevation	Existing Elevation	Cut (ft)
E6	104.0′	104.1′	0.1
F6	104.0′	104.8′	0.8
E7	103.6′	103.6′	0.0
F7	104.2′	104.2′	0.0

bcf of cut = Sum of cut at intersections/Number of intersections x Area

= (0.1 + 0.8 + 0 +0) / 4 x 2500

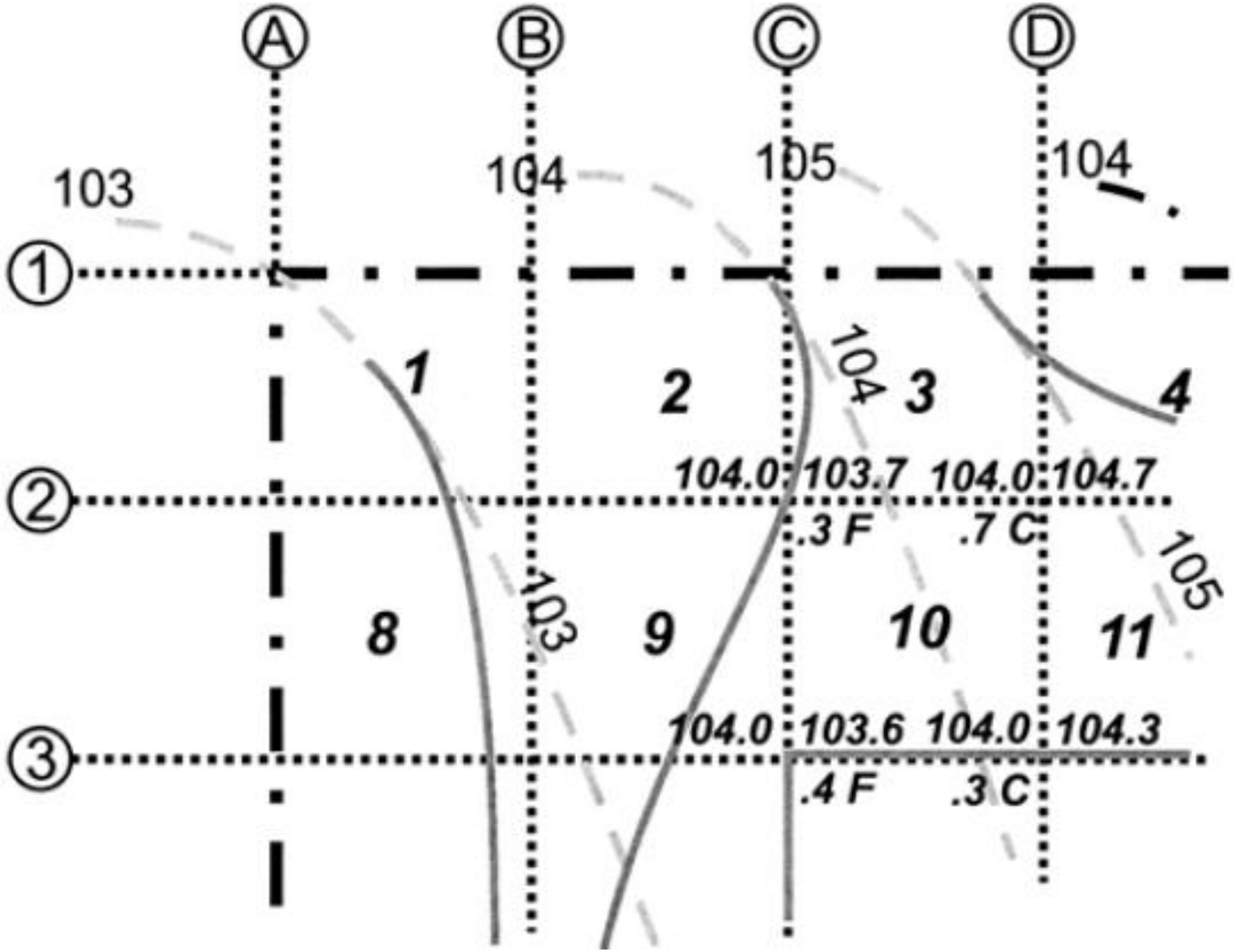
= 563 bcf

This amount is then entered in the cut column of the cut and fill worksheet (Figure 9.16)

Cross-Section Method

- Cuts and fills in same grid
  - Divide grid into separate cut and fill areas

Example 9.6


A
B
C
D
1
2
3
103
104
105
104
104.0 103.7 104.0 104.7
.3 F .7 C
104.0 103.6 104.0 104.3
.4 F .3 C
1
2
3
4
8
9
10
11
104
103
105

Example 9.6 Cut and Fill in the same grid

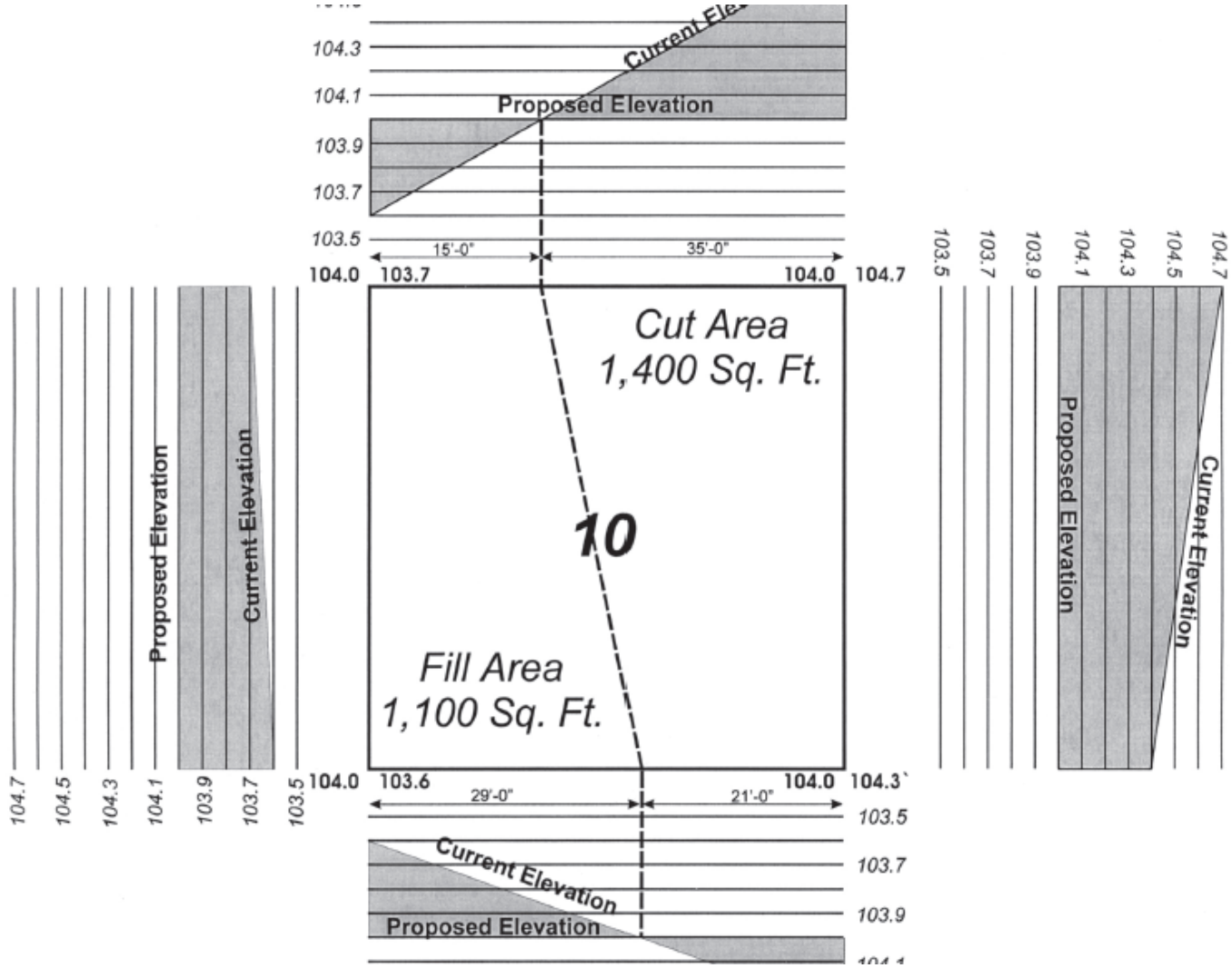
Using Grid 10 from Fig 9.11, Determine the cut and fill quantity

Along line 2, between lines C & D, there is a point where there is no change in elevation. This point is found first.
Total change in elevation (C – D) = 0.3’ + 0.7’ = 1.0’
Change in elevation per foot of run (C – D) = 1.0/50 = 0.02 per foot of run
Thus distance from C2 = 0.3/0.02 = 15’ (Point of no change in elevation)

Along line 3
Total change in elevation (C – D) = 0.4 + 0.3 = 0.7
Change in elevation per foot run (C – D) = 0.7/50 = 0.014 per foot of run
Thus, distance from C3 = 0.4/0.014 = 29’ (Point of no change in elevation)

Having this information , Grid 10 is divided into two distinct grids: One for cut and one for fill (See Fig 9.12: How the grid would be divided)

For further calculation, you can divide the area into rectangles/triangles



Cut and Fill in the Same Grid (Grid 10)

Figure 9.12

Example 9.6 Cut and Fill in the same grid

Using Grid 10 from Fig 9.12, Determine the cut and fill quantity

Fill area (Rectangle) = 15’ x 50’ = 750 sf
Fill area (Triangle) = 1/2x14 x 50 = 350 sf
Total fill area = 750 + 350 = 1100 sf
Fill = (0.3 + 0.4 + 0 + 0) / 4 x 1100 = 195 ccf of fill

The area of cut = Area of grid – area of fill
= 2500 – 1100 = 1400 sf
Cut = (0.7 + 0.3 + 0 + 0) / 4 x 1400 = 350 bcf of cut

These cuts and fills are entered into the cut and fill columns of the worksheet (Fig. 9.16)

Cross-Section Method

- Always keep cuts and fills separate
  - Need to take into account shrinkage
    - Cuts are in bank cubic yards
    - Fills are in compacted cubic yards
- When dealing with pavement, the finish grade is below the base under the pavement

Example 9.7 Cut and Fill

Occasionally when Grid is divided, a portion of the grid will neither be cut or fill. Now considering Grid 3 and Figure 9.13, the change from fill to cut occurs on line 2 between C and D.
Total change in elevation (C – D) = 0.3’ + 0.7’ = 1.0’
Change in elevation per foot of run (C – D) = 1.0/50 = 0.02 per foot of run
Thus distance from C2 = 0.3/0.02 = 15’ (point of no change in elevation)

Fill area = ½ x 15 x 50 = 375 sf
Cut area = ½ x 35 x 50 = 875 sf
Remaining area = 2500 – 375 -875 = 1250 sf (Fig 9.14 shows no cut or fill)

Fill = (0.3 + 0 + 0) / 3 x 375 = 38 ccf of fill

Cut = (0.7 + 0 + 0) / 3 x 875 = 204 bcf of cut

Fig 9.14 is the plan of Grid 3 with the area of no cut and fill shown
Fig 9.15 is the entire plan with the area of no cut and fill shown
Fig 9.16 is the completed cut and fill worksheet for the entire plot.

Example 9.8 Cut with Paving

Using Figure 9.17 and 9.18, determine cut for Grid 9.

In Fig 9.18, the top of the rough grade is 0.50 foot below the top of pavement.

Point	Top of Pavement	Top of Rough grade	Existing elevation	Cut (ft)
C2	100.9	100.4	101.8	1.4
D2	101.0	100.5	101.9	1.4
C3	100.7	100.2	101.4	1.2
D3	100.8	100.3	101.6	1.3

Cut = (1.4 +1.4 + 1.2 +1.3) / 4 x 2500 = 3,313 bcf = 123 bcy

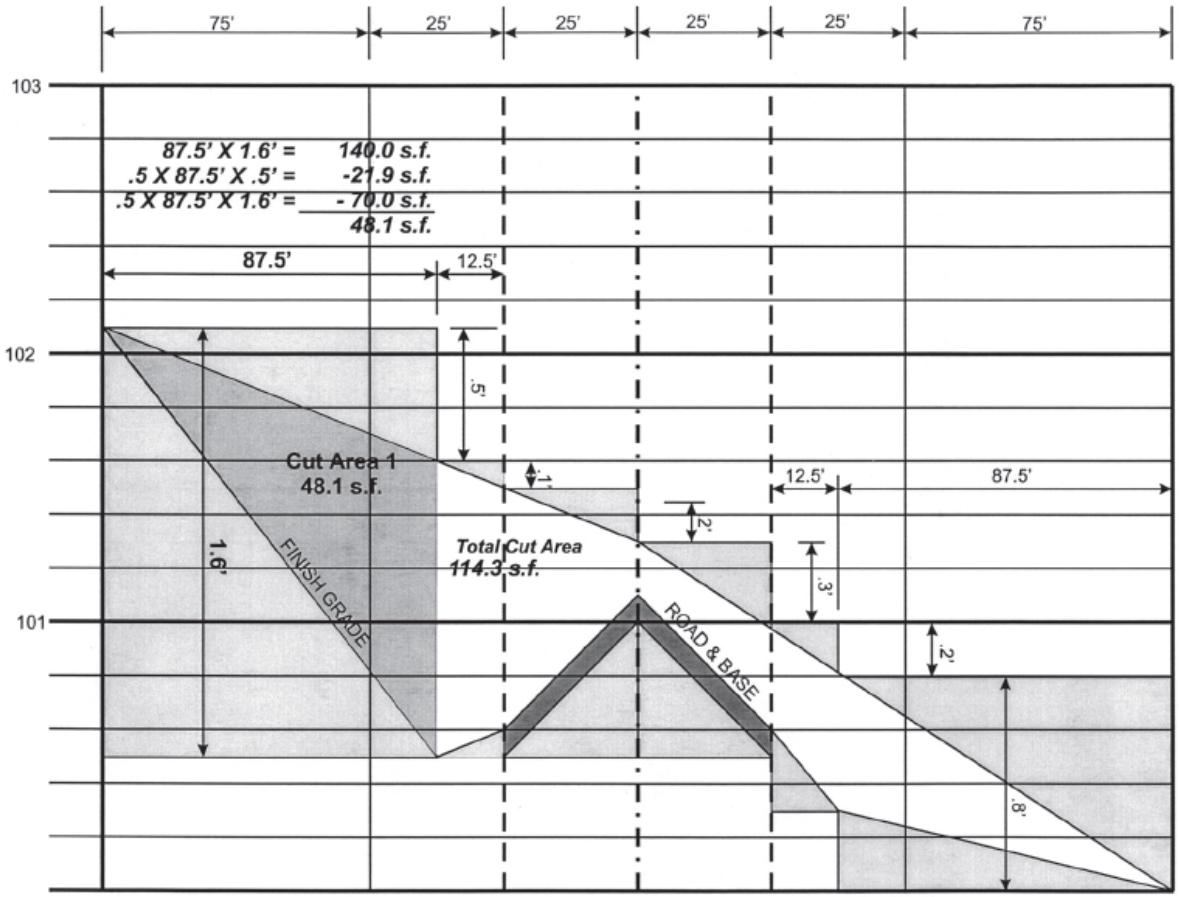
Average End Area

- Used for roads

- Steps
  - Draw profiles for stations
  - Calculate cuts and fills for each station
  - Calculate cuts and fill between stations

Average End Area

- Cut = Distance $(Cut_1 + Cut_2)/2$
- Fill= Distance $(Fill_1 + Fill_2)/2$



Cut Area Calculation at Station 00 + 00

Figure 9.21

Example 9.9 Cut and Fill b/w Station 00+00 and Station 00+75

Profile stations 00+00 and 00+75
Figure 9.21 and Figure 9.22 show the profile for station 00+00
Calculations (partial demonstration)
87.5 x 1.6 = 140.0
1/2x87.5 x 0.5 = - 21.9
½ x 87.5 x 1.6 = - 70.0
= 48.1 sf
Figure 9.23 shows the cut area at station 00+00 = 114.3 sf
" " " station 00+75 = 232.4 sf
These amounts are average and then multiplied by 75 ft

Cut = (114.3 + 232.4) / 2 x 75
= 13,001
= 482 bcy

Example 9.10 Perimeter

See Fig 9.24

Perimeter = 85+25+15+35+30+10+30+10+40+60

= 340 ft

Example 9.11 Building Area

See Fig 9.24

Basic Area 100′ x 60	=	6,000
Bottom recess 10′ x 30′	=	-300
Top recess 15′ x 25	=	-375
Net building area	=	5,325 sf

Topsoil Removal

- From
  - Building
  - Walks
  - Roadways
  - Parking areas

Example 9.12 Topsoil Removal

See Fig 9.25

Building enlarged by 5′ for slope = 5′

Topsoil removal = 9″ thick

Quantity of topsoil removed and stockpiled = ?

Quantity of topsoil to be removed = 110 x70 x .75

= 5,775 bcf

= 214 bcy

*Example 9.13 Topsoil - Equipment and labor cost

See Fig 9.26 & 9.27

1-cy front-end loader is selected for top soil removal

Production rate of front-end loader = 24 bcy per hour

Mobilization time = 2.5 hours

Operating cost of loader= $ 11.35 per hour

Operator cost = $ 17.75 per hour

Quantity of topsoil = 214 bcy

Estimate the no. of hours and the cost to strip the topsoil = ?

Hours for the quantity = 214/24 = 8.92

Total hours =8.92 + 2.5 = 11.4 hours

Equipment cost = 11.4 x 11.35 = $129

Labor cost = 11.4 x 17.75 = $202

Total cost = 129 + 202 = $ 331

General Excavation Factors

- Building size
- Footing projection beyond wall
- Working space
- Existing grade (elevations)
- Soil type (see borings or soils report)
- Slope of excavation or shoring
- Depth of excavation

Angles of Repose

	Angle		
Material	Wet	Moist	Dry
Gravel	15-25	20-30	24-40
Clay	15-25	25-40	40-60
Sand	20-35	35-50	25-40

Basement Excavation

Average length (or width) of cut
= Horizontal distance of slope/2
+ Working space
+ Footing projection
+Length (or width)
+ Footing projection
+ Working space
+ Horizontal distance of slope/2

Basement Excavation

Volume
=Average length of cut
x Average width of cut
x depth of cut

Continuous Footings

Volume
=Length of footing
x Average width of cut
x Depth of cut

Spread Footings

Volume
=Average length of cut
x Average width of cut
x depth of cut

Example 9.14 Basement Excavation

See Fig 9.32, 9.33 & 9.34

1. Exterior dimensions = 26′ 1″ x 32′ 4″
2. Footing projects from wall = 1′
3. Workspace from edge of footing = 1′
4. Depth of cut after removing the 9″ topsoil = 8′
5. Check soil boring, slope = 2:1 (2′ vertical and 1′ horizontal)
6. Basement slab thickness = 1′

Average Width of cut = 2 +1 + 28′ 1″ + 1 +2 = 34′ 1″

Average Length of cut = 2 +1+ 34′ 4″ + 1 + 2 = 40′ 4″

General excavation = 34′ 1″ x 40′ 4″ x 8 = 34.083 x 40.333 x 8 = 10,997 bcf

= 407 bcy

Continue…

Example 9.14 Basement Excavation

See Fig 9.32, 9.33 & 9.34
Backhoe = 1 cy bucket
Mobilization = 2 hrs
Rate of work for backhoe = 55 bcy per hour
Equipment cost = $16.75 per hour
Operator cost = $19.75 per hour

Equipment hour = 407/55 + 2 = 7.4 +2 = 9.4 hours
Equipment cost = 9.4 x 16.75 = $157
Labor cost = 9.4 x 19.75 = $186
Total cost = 157 + 186 = $343
Calculate hauling cost by using 7 cy truck ?
Assume swell = 30%
Lcy = 407 x 1.3 = 529 lcy
Required loads = 529/7 = 76 loads

Example 9.15 Continuous Footing Excavation

See Fig 9.32, 9.35 sketch of continuous footing

Slope = 1.5 : 1 (1.5 vertical to 1 horizontal) (4 : 1.5 :: x : 1)

Building perimeter = 340

Special Excavation

- Hand excavation
  - Small trenches
  - Small cuts below general excavation

Backfilling

- Factors
  - How will material be transported?
    - Wheelbarrows
    - Front-end loaders
    - Etc.
  - Is compaction required?

Backfilling

- Volume = Volume of excavation – Volume of building or footing

- Needs to exclude topsoil

Excess or Borrow

- Excess
  - Disposed on site or
  - Hauled off
- Borrow
  - Source
- Need to take into account shrink when importing and swell when transporting

Landscaping

- Fertilizer
- Sod
- Trees and shrubs
- Maintenance period

Dewatering

- Factors
  - Elevation of water table
    - Varies through the year

Piles

- Usually subcontracted out

- Cost based on
  - Design
  - Conditions of use
  - Types of soil
  - Depth

Asphalt Paving

- Layers are bid by the ton

Asphalt Paving

- Subgrade preparation
- Subdrains
- Soil sterilization
- Insulation course
- Subbase course
- Base courses
- Prime and tack coats
- asphalt

Thank You